Predicting Product Review Sentiment Using Machine Learning and LLMs

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2026 Spring

How do traditional ML models compare to LLMs at Sentiment Analysis Tasks

ML for Sentiment Analysis

- Sentiment analysis is used at scale for tasks like:
  - Summarizing product reviews
  - Analyzing user feedback
  - Product marketing
- We want to better understand:
  - How traditional ML models compare to LLMs
  - What balance might exist between performance and cost
- We will build upon prior work in this area that found LLMs to be superior to tradition ML methods [1]

Methods - Data Preprocessing

Dataset & Labels

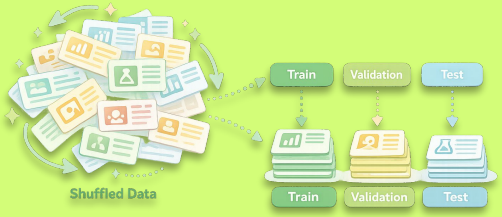
- Amazon Product Reviews dataset (Kaggle)
- We converted ratings into binary sentiment:
  - Positive (1): 4–5 stars
  - Negative (0): 1–2 stars
  - We removed neutral (3-star) reviews.

Sampling & Split

- We selected 50,000 reviews, shuffled the data, and split into:
- 40k train, 5k validation, 5k test

Data Formats

- Raw text for LLM evaluation
- Preprocessed text for traditional ML

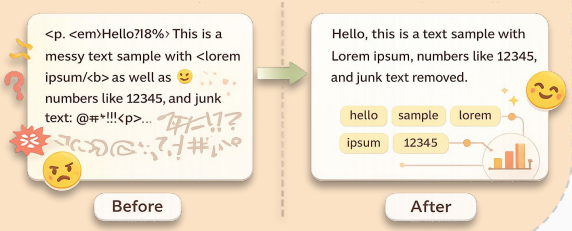


Preprocessing (ML Only)

- Lowercasing, HTML removal, punctuation/number removal
- Stopword removal/lemmatization

Why This Matters

- Reduces noise and standardizes input for ML
- Preserves full context for LLM comparison



Methods - Traditional ML Models

- Models
  - SVM (Linear SVC), Logistic Regression, and Random Forest
- Preprocessing
  - Used preprocessed dataset (cleaned text)
- Training Setup
  - Trained on training set
  - Used validation set for checking performance
- Evaluation
  - Measured Accuracy, Precision, Recall, and F1-score
  - Used classification report for detailed results

Methods - Traditional ML Models

- Sentiment analysis is used at scale for tasks like:

Methods - LLMs

- Models
  - GPT 5 Mini and GPT 5 Nano
    - Selected GPT 5 Mini for full run due to API limits
- Prompt Strategy
  - Opted for a “zero-shot” user prompt including the review
  - Response either 0 for negative or 1 for positive
  - Single review per conversation, avoiding continuous context
- API Use
  - Registered an API key with $5, spent only $0.73
  - Leveraged LangChain for rate-limit aware API wrapper
  - Conducted token and cost estimates prior to full run of the dataset
- Evaluation
  - Used only the test set, but unprocessed with full text

Results - SVM & Logistic Regression

SVM

92%

Accuracy

94%

Precision

Tied for best precision score

97%

Recall

95%

F1 Score

Logistic Regression

91.5%

Accuracy

92%

Precision

99%

Recall

Best overall recall score

95%

F1 Score

Results - Random Forest & LLM

Random Forest

85%
Accuracy

85%
Precision

85%
Recall

80%
F1 Score

GPT 5 Mini

98%
Accuracy

Best overall accuracy score

94%
Precision

Tied for best precision score

97%
Recall

96%
F1 Score

Best overall F1 score

Conclusion

- Overall our results show:
  - LLM approach was superior based on F1 score
    - F1 score matters the most given the imbalanced dataset
  - Traditional ML models SVM & Logistic Regression closely followed LLM F1 score
  - Random Forest performed the worst over all the metrics
  - LLMs are relatively inexpensive to use

Limitations

Dataset Constraints

- Imbalanced data (~84% positive, 16% negative)
- Binary labels remove real-world nuance

Model Comparison Fairness

- ML used preprocessed TF-IDF text
- LLM used raw text
- Comparison is informative but not perfectly equivalent

LLM Constraints

- API rate limits and token limits affected model choice
- Used GPT-5 mini with a zero-shot prompt
- Results vary with different prompts or scale

Scope & Generalization

- Limited set of models tested
- Based on one dataset, may not generalize to other domains

Trade-offs

- Dataset
  - The dataset was simplified to accommodate simpler ML models using binary classification
- Traditional ML Methods
  - Pros
    - Fast to train and run
    - Easy to implement and reproduce
  - Cons
    - Requires preprocessing (cleaning text, TF-IDF)
    - Struggles with complex language
- LLMs
  - Cons
    - Requires continuous spend on API credits
    - Challenging to abide by API limits at large scale
    - Dependent on third party services being available
  - Pros
    - Limits increase with more usage
    - Token caching greatly reduces overall use and cost

Thank you!

Questions?

Project GitHub Repository
https://github.com/KyKri/cs561-project-group-2/

- [1] P. S. Ghatora, S. E. Hosseini, S. Pervez, M. J. Iqbal, and N. Shaukat, “Sentiment analysis of product reviews using machine learning and pre-trained LLM,” Big Data and Cognitive Computing, vol. 8, no. 12, p. 199, Dec. 2024, doi: 10.3390/bdcc8120199.
- [2] A. Rumi, “Amazon Product Reviews,” Kaggle, 2023. [Online]. Available: https://www.kaggle.com/datasets/arhamrumi/amazon-product-reviews. Accessed: Mar. 17, 2026.
- [3] OpenAI, “API Pricing,” OpenAI, [Online]. Available: https://openai.com/api/pricing/. [Accessed: Apr. 7, 2026].